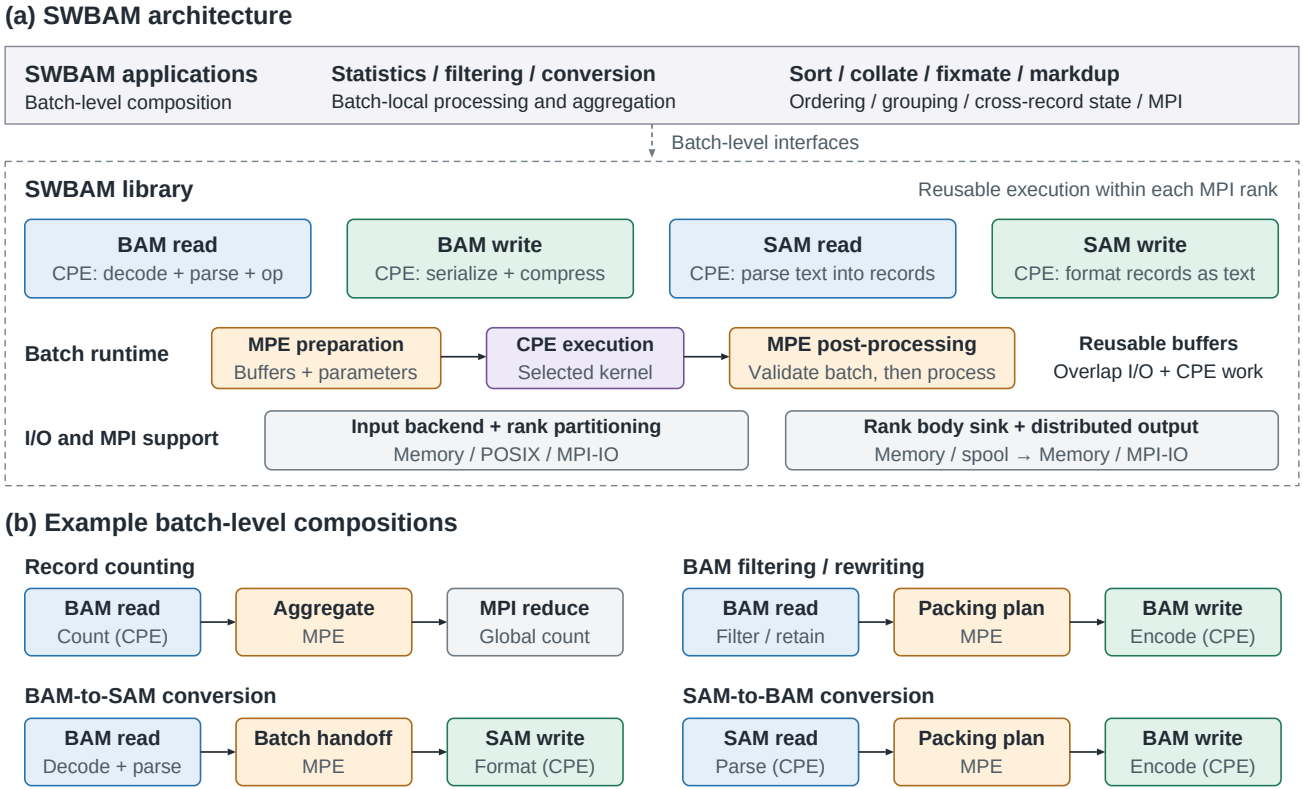
(a) SWBAM architecture
SWBAM applications
Batch-level composition
Statistics / filtering / conversion
Batch-local processing and aggregation
Sort / collate / fixmate / markdup
Ordering / grouping / cross-record state / MPI
Batch-level interfaces
SWBAM library
Reusable execution within each MPI rank
BAM read
CPE: decode + parse + op
BAM write
CPE: serialize + compress
SAM read
CPE: parse text into records
SAM write
CPE: format records as text
Batch runtime
MPE preparation
Buffers + parameters
CPE execution
Selected kernel
MPE post-processing
Validate batch, then process
Reusable buffers
Overlap I/O + CPE work
I/O and MPI support
Input backend + rank partitioning
Memory / POSIX / MPI-IO
Rank body sink + distributed output
Memory / spool → Memory / MPI-IO
(b) Example batch-level compositions
Record counting
BAM read
Count (CPE)
Aggregate
MPE
MPI reduce
Global count
BAM filtering / rewriting
BAM read
Filter / retain
Packing plan
MPE
BAM write
Encode (CPE)
BAM-to-SAM conversion
BAM read
Decode + parse
Batch handoff
MPE
SAM write
Format (CPE)
SAM-to-BAM conversion
SAM read
Parse (CPE)
Packing plan
MPE
BAM write
Encode (CPE)